No. of Functional Features Included in the Solution

Date	11 July 2026
Team ID	SWTID-2026-1298
Project Name	Smart Lender
Maximum Marks	5 Marks

Functional Features:

S.No	Functional Feature	Description	Module	Status	Contribution
1	Dataset Loading	Imports the loan applicant dataset from CSV and prepares it for processing.	Data Acquisition	Done	High
2	Data Cleaning & Transformation	Handles missing values, converts categorical attributes into machine-readable format and prepares clean data.	Data Preparation	Done	High
3	Exploratory Data Analysis	Generates statistical summaries and visualizations to understand applicant patterns and relationships.	Data Analysis	Done	Medium
4	Machine Learning Model Development	Builds and trains multiple classification algorithms including Decision Tree, Random Forest, KNN and XGBoost.	Model Training	Done	High
5	Performance Comparison	Evaluates all trained models using accuracy score and selects the best-performing classifier.	Model Evaluation	Done	High
6	Loan Eligibility Prediction	Accepts applicant information and predicts whether the loan is likely to be approved or rejected.	Prediction Engine	Done	High
7	Web Application Development	Develops a responsive Flask-based interface for user interaction.	Web Interface	Done	Medium
8	Model Deployment	Integrates the trained Random Forest model with the Flask application for real-time prediction.	Deployment	Done	Medium

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	7
Additional / Nice-to-Have Features	1
Features Tested & Verified	8

Feature Category Breakdown:

Category	Count	Included Features	Category
User Interface	1	Loan Eligibility Input Form	User Interface
Data Processing	2	Dataset Loading, Data Cleaning	Data Processing
Machine Learning	3	Model Training, Evaluation, Prediction	Machine Learning
Deployment	1	Flask Integration	Deployment
Storage	1	CSV Dataset and Pickle Model	Storage
Authentication	0	Not Imlemented	

Key Functional Highlights

- Import and preprocess loan application data.
- Handle missing values using statistical techniques.
- Encode categorical features for machine learning.
- Train four different machine learning algorithms.
- Compare model accuracy and choose the best model.
- Save the trained model using Pickle.
- Build an interactive Flask web application.
- Predict loan approval instantly based on user input.